
The 21st Century Lab Notebook

Quick Start Guide

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QL21 • Quick Start Guide

QL21 is a lightweight framework for documenting research projects in a reproducible and transparent way. It uses Git and open text-based formats so that project records remain accessible over time. No programming experience is required.

1. Install the tools

Layer 1 — Required formats & version control

Format / Tool	Purpose
Markdown	Plain text format for logs, notes & reports
YAML	Metadata & frontmatter for all documents
BibTeX	Reference format for bibliography
LaTeX	Format for academic writing & final reports
Git	Local version control & history

The tools listed below are suggestions rather than requirements. If your institution already provides equivalent solutions, you can continue using them. The main goal is to work with open formats and maintain a clear project history.

Layer 2 — Recommended

Tool	Purpose
Obsidian	Write & organise Markdown docs
VS Code	Editor for YAML, Markdown & scripts
Zotero	Reference manager
Pandoc	Convert Markdown to PDF/Word

Layer 3 — Optional (AI)

AI tools can be useful for drafting, summarising, reviewing, or organising research materials. However, responsibility for the accuracy and interpretation of the content always remains with the researcher. For confidential or sensitive work, consider running models locally using Ollama. For confidential research, use Ollama locally.

2. Start a new project (30 min)

The purpose of the initial setup is simply to establish a consistent structure for project documentation. Most files can be completed gradually as the project develops.

```
git clone https://github.com/username/my-project.git
cd my-project
mkdir 00-governance 01-literature 02-framework 03-pilot 04-outputs
↪ 05-journal
```

Copy the templates from 02-framework/templates/. Fill in the governance manifests in 00-governance/. First commit:

```
git add .
git commit -m "feat: initialise project structure v0.1.0"
git push -u origin main
```

3. Daily workflow

The workflow is designed to leave a lightweight record of each working session. This makes it easier to resume work after a break, review decisions, and share progress with collaborators.

When	What	Where
Project start	Copy TPL-PROJ.md, rename, fill in	03-pilot/
Start of session	Copy TPL-LOG.md, rename, fill in	03-pilot/logs/
New insight	Copy TPL-ZN.md, rename, fill in	05-journal/
End of week	Copy TPL-RPT.md, rename, fill in	03-pilot/
End of session	Commit & push	—

Naming convention: {PREFIX}-{YYYYMMDD}T{HHmmss}-{AUTHOR}-1.00.md

End every session with:

```
git add .  
git commit -m "docs: add LOG session YYYY-MM-DD"  
git push
```

4. Commit prefixes

Prefix	Use for
feat	New template or artefact
docs	Log, note, report
fix	Correction
chore	Maintenance
refactor	Renaming or restructuring

5. AI assistance (optional)

Attach these files at the start of every AI session: -user-manifest.yaml -project-manifest.yaml
-governance-manifest.yaml - Last RPT from 05-journal/

Use local Ollama for confidential research.

6. Project closure

When a project reaches a stable final state, create a version tag to identify the release that supports reports, publications, or archived outputs.

```
git tag -a v1.0.0 -m "Project closure: final version"  
git push origin v1.0.0
```

→ Full guide: D4 Best Practices Guide

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